

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: DEEP LEARNING

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO2: Analyze Machine Learning and Deep Learning models by evaluating neural network architectures, representation learning, activation functions, unsupervised learning techniques, and their real-world applications. (L4)

CO Weightage: 15%

Assessment Tool Weightage: 20%

QUESTIONS:

Total Marks: 20

1. Differentiate between Machine Learning and Deep Learning with respect to architecture, data requirements, feature extraction and applications. Give two real-world applications of each.
2. Explain the concept of Representation Learning. Discuss how the width and depth of a neural network influence its learning capability, computational complexity and model performance.
3. Compare ReLU, Leaky ReLU (LReLU) and ELU (Exponential Linear Unit) activation functions. Explain their mathematical behavior, advantages, disadvantages and suitable use cases.
4. Explain the concept of unsupervised learning in neural networks. Describe the working principle of Restricted Boltzmann Machines (RBMs) and Autoencoders and compare their objectives and applications.

Code of Conduct:

I Haresh Kumar N L (192425009) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

1) Introduction:-

→ AI is a broad field that enables machines to perform tasks that normally require human intelligence. Two important branches of AI are ML and DL.

Difference between ML and DL:-

Architecture:-

ML:- uses simple algorithms such as Decision Trees, Naive Bayes, Random Forest, SVM, and Linear Regression. usually consists of one or a few processing layers.

DL:- uses ANN with many hidden layers, known as DNN, CNN, RNN, and Transformers.

Data Requirements:-

ML:- Performs well even with small or medium-sized datasets.

DL:- Requires very large datasets to achieve high accuracy and better generalization.

Feature Extraction:-

ML:- Features must be manually selected and engineered by domain experts before training.

DL:- Automatically extracts important features from raw data without manual intervention.

Hardware Requirement:-

ML:- Can be trained efficiently on CPUs.

DL:- Requires powerful GPUs or TPUs for faster computation and training.

Interpretability :-

ML:- Easier to understand and explain model decisions.

DL:- often considered a "black-box" model because internal decisions are difficult to interpret.

Performance:-

ML:- Works well on structured data such as spreadsheets and databases.

DL:- Excels in unstructured data like images, videos, speech, and natural language.

Architecture Comparison:-

ML Architecture:-

Input Data $\rightarrow$ Data Preprocessing - Feature Engineering $\rightarrow$ ML $\rightarrow$ Prediction.

DL Architecture:-

Input layer $\rightarrow$ Hidden layer 1 $\rightarrow$ Hidden layer 2 $\rightarrow$ $\rightarrow$ Output layer.

Advantages:-

ML:-

- $\rightarrow$ Requires less training data.
- $\rightarrow$ Faster model training.

DL:-

- $\rightarrow$ Automatically learns features.
- $\rightarrow$ Reduces manual feature engineering.

Limitations:-

ML:-

- $\rightarrow$ Requires manual feature extraction.
- $\rightarrow$ Less effective for high complex problems.

DL:-

- $\rightarrow$ Needs very large datasets.
- $\rightarrow$ Long training time.

2) Introduction:-

Representation Learning is a technique in DL where a neural network automatically learns useful features from raw data without manual feature extraction. Unlike traditional ML, where experts manually select features, DL Learning learns these features through multiple hidden layers. This enables the model to identify complex patterns in images, text, speech, and other types of data.

Representation Learning:-

Representation learning helps neural networks transform raw input into meaningful information. Each hidden layer learns different levels of features.

Advantages:-

- Automatic Feature extraction
- Higher prediction accuracy
- Handles complex and unstructured data.
- Reduces human effort.

Width of a Neural Network:-

The width of a neural network refers to the number of neurons in each hidden layer.

Depth of a Neural Network:-

The depth of a neural network refers to the number of hidden layers.

Comparison of width and Depth:-

Definition:-

Width:-

Number of neurons in a hidden layer

Depth:-

Number of hidden layers

Learning:-

Width:-

learns more Features simultaneously

Depth:-

learns hierarchical Features

Complexity:-

Width:-

Increases memory and computation

Depth:-

Increases training time

Performance:-

Width:-

Better Feature Learning

Depth:-

Better abstraction accuracy.

Applications:-

→ Image Recognition

→ NLP

→ speech Recognition

→ Healthcare

3) Introduction:-

Activation Functions are an essential Part of ANN. They Introduce non-linearity into the Network, allowing it to learn complex relationships between I/P and o/P.

ReLU (Rectified Linear unit):-

Mathematical Behavior:-

The ReLU Function is defined as:

$$f(x) = \max(0, x)$$

→ If $x > 0$, the output is x

→ If $x \leq 0$, the output is 0.

Advantages:-

→ Simple and computationally efficient.

→ Faster training of DNN.

Disadvantages:-

→ Suffers From the dying ReLU problem

→ Output zero For all negative values.

Leaky ReLU (LReLU):-

Mathematical Behavior:-

Leaky ReLU is defined as:

$$f(x) = \max(\alpha, x)$$

→ If $f(x) = x$, $x > 0$

→ If $f(x) = \alpha x$, $x \leq 0$

Advantages:-

- Solves the dying ReLU problem.
- Allows neurons to remain active.
- Improves gradient flow during training.
- Better Learning Performance in some deep networks.

Disadvantages:-

- The value of α must be selected carefully.
- Slightly more computationally expensive than ReLU.

ELU (Exponential Linear Unit)

Mathematical Behavior:-

The ELU function is defined as:

- $f(x) = x$, if $x > 0$
- $f(x) = \alpha (e^x - 1)$, if $x \leq 0$

Advantages:-

- Reduces vanishing gradient problem.
- Produces Faster convergence.
- Mean activations become closer to zero.
- Often provides better accuracy than ReLU.

Disadvantages:-

- Computationally more expensive because of the exponential calculation.
- Slower than ReLU during training.

4) Introduction:-

unsupervised learning is a type of ML in which the model learns patterns from unlabeled data. unlike supervised learning, it does not require target output or class labels. The neural network analyzes the input data, discovers hidden structures, and groups similar data automatically. unsupervised learning is widely used for clustering, dimensionality reduction, feature learning, anomaly detection, and data compression.

Restricted Boltzmann Machine (RBM)

A Restricted Boltzmann machine is a stochastic neural network used for feature learning and dimensionality reduction. It consists of two layers.

- visible layer
- Hidden layer

Advantages:-

- Learning hidden features automatically.
- Reduces data dimensions.
- useful for recommendation systems.
- Can be used for pre-training deep neural networks.

Applications:-

- Feature extraction.
- Collaborative Filtering.
- Dimensionality reduction.

Autoencoder:-

An Autoencoder is a neural network designed to learn an efficient representation of input data by compressing and reconstructing it.

- Encoder
- Latent space
- Decoder

Advantages:-

- Learns meaningful features automatically.
- Performs dimensionality reduction.
- Removes noise from data.
- Improves data compression.

Applications:-

- Image denoising
- Data compression
- Anomaly detection.
- Feature extraction.